Yanzhe Wang

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Education

Princeton University **Sep. 2025 – Jun. 2026 (expected)**
M.S. in Finance *Princeton, NJ, US*

Peking University **Sep. 2021 – Jun. 2025**
B.S. in Mathematics; Minor in Economics *Beijing, China*

Shanghai High School **Sep. 2018 – July 2021**
Shanghai, China

Research Interests

Machine Learning and its intersections with Financial Market, Applied Probabilities, Statistics.

Selected Internship Experiences

Jane Street **Jun. 2024 – Aug. 2024 & Jul. 2025 – Aug. 2025**
Quantitative Trader *Hong Kong, China*

- Projects: Using metadata to predict return of options and trading volume of equities in Asian markets.
- Mock Trading: Practice trading skills in different mock trading games.

Tongdeng Capital **Jan. 2024 – Apr. 2024**
Quantitative Researcher *Shanghai & Beijing, China*

- Utilized data from the past ten trading days to forecast the dominant contracts for 73 futures products in the Chinese futures market.
- Independently developed two distinct algorithmic solutions to address this issue: one with optimized parameters for linear regression, and another employing a simple RNN (Recurrent Neural Network) with a custom loss function.
- Enhanced the prediction accuracy; reduced the forecast loss from the existing company algorithm's 2.49% to 1.39%. The model is now in production use by the company.

Lianhai Capital **Aug. 2023 – Sep. 2023**
Quantitative Researcher *Shanghai, China*

- Built from scratch a two-layer GRU weekly frequency factor model for A-shares database and live trading system. Employed a hybrid RNN+NN architecture for short-term return forecasting and utilize heuristic search to randomly select factor combinations from the factor library.
- Trained and analyzed 600 unique factor combinations; back-tested the results and assessed the impact of various hyper parameters, orthogonalization functions, and other model configurations on the model's stock selection and timing capabilities. The best model reached a Sharpe of 3.86.
- Submitted two factor ideas per week that met the company's back-testing criteria.

Tiger Brokers **Jul. 2023 – Aug. 2023**
International Business Development *Beijing, China & New York City, US*

- International Competitive Business Analysis: Conduct comprehensive assessments of the company's securities products across various markets in Asia and the United States. Analyze key features, revenue models and target customer segments. and submit a detailed report outlining the comparative pros and cons of our products, along with strategies for improvement.

Honors and Awards

- Yizheng Scholarship First-Class Award of Peking University 2024
- Academic Excellence Award of Peking University 2022
- Third-class Scholarship of Peking University 2022
- Gold Medal in the 35th China Mathematical Olympiad 2019

Selected Research Experiences

Multi-Agent Reinforcement Learning Trading Platform **Feb. 2024 – Oct. 2024**
Advised by Prof. Yangbo He, Working with Mr. Ji Wang and Mr. Zhihan Liu *Peking University, China*

- Developed a stock trading simulation platform using multi-agent reinforcement learning to optimize trading strategies under different market conditions.
- Implemented noise and intelligent traders, analyzing the effects of trading frequency, order delay, and price fluctuations on strategy returns.
- Utilized Double Deep Q-Networks (DDQN) to enhance decision-making, with experiments highlighting the importance of historical data and optimized price ranges for profitability.

Discrete Diffusion Model **Feb. 2024 – Oct. 2024**
Advised by Prof. Mengdi Wang and Prof. Minshuo Chen, Working with Mr. Hengyu Fu *Princeton University, US*

- Developed and analyzed discrete diffusion models for generating data distributions on finite support, drawing on forward and backward processes with time-dependent transition matrices.
- Validated the models through experiments in tasks such as multi-armed bandit, achieving high similarity between generated and real samples. Proficient in leveraging score matching objectives, error decomposition, and estimation bounds to advance discrete generative methods, while enhancing computational efficiency and distribution accuracy.

Implied Volatility of Black-Scholes model and Corrado-Su model **Jun. 2023 – Dec. 2023**
Advised by Prof. Alan Anderson *New York University, US (remote)*

- Studied the volatility smile of BS model; studied CS model, the deformation of Black-Scholes model proposed by Corrado and Su that introduce implied third-order moments and fourth-order moments and its calculation method.
- Studied the implied volatility performance of two call options, SSE50ETF and CSI300ETF, between 2018 to 2020. Researched on the comparison of the models' ability to predict tomorrow's prices across different cross-sections.
- Formed a more comprehensive view about Chinese option market and the features of its volatility; studied topics including stochastic analysis and CS option pricing model; grew interests in financial derivatives and its pricing.

High-frequency Pricing Model **Nov. 2022 – May. 2023**
Advised by Prof. Ruixun Zhang, Working with Mr. Ruichen Wang *Peking University, China*

- Studied the Micro Price Model in High-frequency Pricing and CJ and VR(q) statistical tests for random walk hypothesis.
- Judged the extent of rejecting the three random walk hypotheses of Mid/Weighted/Micro price by CJ/VR(q) variance ratio; evaluated the estimators by the correlation coefficients that portray the predictive power for future prices within the same quarter under stock and date.
- Demonstrated the optimality of Mid/Weighted model under linearity hypothesis by ML/DL methods. Constructed a new high-frequency spread pricing model that outperforms the three existing models in statistical tests by ML methods.
- My contributions: Proposed a methodology capable of evaluating the ability of three pricing models to predict future price movements and implemented the full analytical process; completed part of ML code for demonstrating the optimality of Mid/Weighted model.

Activity Experiences

School Floorball Team Player **Dec. 2023 – Jul. 2025**

Member of Peking University Orientation Association **Sep. 2021 – Jun. 2025**
Individual: 5th place; Team: 2nd place *2021 "Peking University Cup" Orientation Race*

Standardized Language Tests

- TOEFL iBT, Total 107 (Reading 29, Listening 28, Speaking 24, Writing 26)
- GRE General Test, Total 326 (Verbal Reasoning 156, Quantitative Reasoning 170)